English

(1015CMD303361230045)

ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

Test Pattern

NEET (UG)

MINOR

16-07-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE-MEF-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

Test Booklet Code

This Booklet contains **28** pages.

E5

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II). 50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720**.
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : ______________________

Form Number : in figures ______________________

: in words ______________________

Centre of Examination (in Capitals) : ______________________

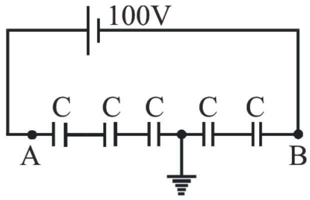
Candidate's Signature : ______________ **Invigilator's Signature :** ______________

Your Target is to secure Good Rank in Pre-Medical 2024

Topic : WORK, POWER, ENERGY, THERMAL PHYSICS, MAGNETISM (TILL TAUGHT), CAPACITOR, [BACK UNIT] : CURRENT ELECTRICITY

SECTION - A (PHYSICS)

1. Find potential at A and B in the given circuit–



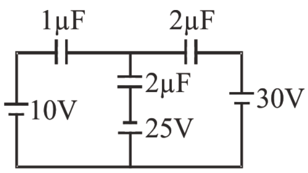
(1) 40 V, 0 V

(2) 50 V, 0 V

(3) 60 V, –40 V

(4) 40 V, –10 V

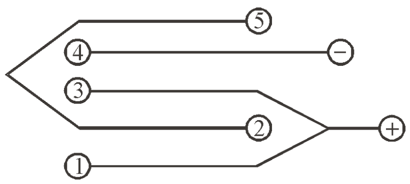
2. In the given circuit find out the charge on 1μF capacitor. (Initially they are uncharged) :-



(1) 52 μC (2) 58 μC

(3) 12 μC (4) 6 μC

3. Five conducting plates 1,2,3,4 & 5 are fixed parallel to and equidistant from each other as shown in figure if 'd' is distance between two successive plates and A is area of either face of each plate. Then find net capacitance of system:-



(1) $\frac{7}{3}\frac{\epsilon_0 A}{d}$

(2) $\frac{5}{3}\frac{\epsilon_0 A}{d}$

(3) $\frac{3}{7}\frac{\epsilon_0 A}{d}$

(4) $\frac{3}{5}\frac{\epsilon_0 A}{d}$

4. A parallel plate capacitor of capacitance C is connected to a cell of emf V. A dielectric slab of dielectric constant K, that can just fill the air gap of capacitor, is now inserted in it. Which of the following is incorrect?

(1) The energy stored in the capacitor increases K times

(2) The change in energy stored is $\frac{1}{2}CV^2(K-1)$

(3) The charge on the capacitor remain same as before

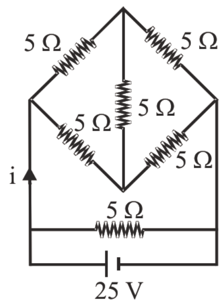
(4) The potential difference between the plates remain unchanged.

5. The two parallel plates of a condenser have been connected to a battery of 300 V and the charge collected at each plate is 1 μC. The energy supplied by the battery is :

(1) 6×10^{-4}J (2) 3×10^{-4}J

(3) 1.5×10^{-4}J (4) 4.5×10^{-4}J

6. The value of current i for the given circuit is :-



(1) 10 A (2) 5 A (3) 2.5 A (4) 20 A

7. A potentiometer is an accurate and versatile device to make electrical measurements of E.M.F. because the method involves :-

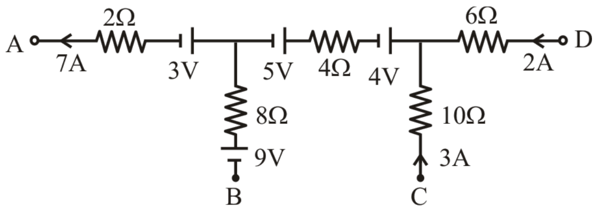
(1) Potential gradients

(2) A condition of no current flow through the galvanometer

(3) A combination of cells, galvanometer and resistances

(4) Cells

8. In a portion of some large electrical network, current in certain branches are known as shown in figure :-

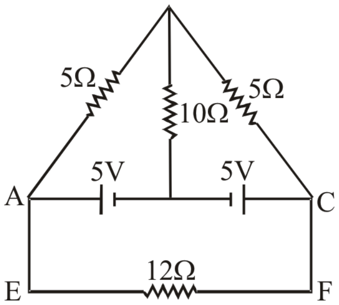


The value of $V_A - V_C$ is

(1) 76 V (2) – 76 V

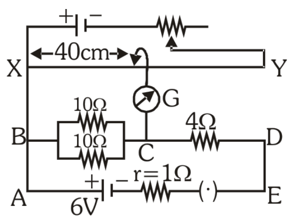
(3) –58 V (4) –52 V

9. In the circuit of adjoining figure the current through 12Ω resister will be :-



(1) 1 A (2) $\frac{1}{5}$ A (3) $\frac{2}{5}$ A (4) 0 A

10. In the following circuit the potential difference between the points B and C is balanced against 40 cm length of potentiometer wire. In order to balance the potential difference between the points C and D, where should jockey be pressed?



(1) 32 cm (2) 16 cm

(3) 8 cm (4) 4 cm

11. An uncharged capacitor of capacitance 4μF, a battery of emf 12 volt and a resistor of 2.5 MΩ are connected in series. The time after which $v_c = 3v_R$ is [take $\ell n2 = 0.693$]

(1) 6.93 sec. (2) 13.86 sec.

(3) 20.52 sec. (4) none of these

12. If you have several 2.0μF capacitors, each capable of withstanding 200 volts without break down.
Then find minimum no. of capacitors required in a combination for equivalent capacitance 0.40 μF and applied voltage of 1000V.

(1) 7 (2) 5 (3) 9 (4) 6

13. A potentiometer having a wire 10 m long stretched on it is connected to a battery having a steady voltage. A leclanche cell gives a null point at 750 cm. If the length of potentiometer wire is increased by 100 cm, find the new position of null point :-

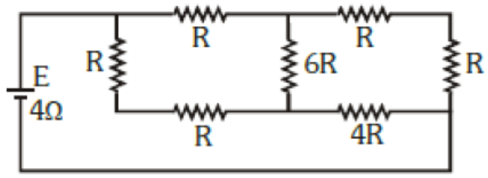
(1) 750 cm

(2) 675 cm

(3) 825 cm

(4) 900 cm

14. A battery of internal resistance 4Ω is connected to the network of resistance as shown. In order that the maximum power can be delivered to the network, the value of R in Ω should be :-



(1) $\frac{4}{9}$ (2) 2 (3) $\frac{8}{3}$ (4) 18

15. A battery of internal resistance 2Ω is connected to a variable resistor whose value can vary from 4Ω to 10 Ω. The resistance is initially set at 4Ω. If the resistance is now increased then :

(1) power consumed by it will decrease

(2) power consumed by it will increase

(3) power consumed by it may increase or may decrease

(4) power consumed will first increase decrease

16. A closely packed coil having 1000 turns has an average radius of 62.8 cm. If current carried by the wire of the coil is 1 A, the value of magnetic field produced at the centre of the coil will be (permeability of free space = $4\pi \times 10^{-7}$ H/m) nearly:

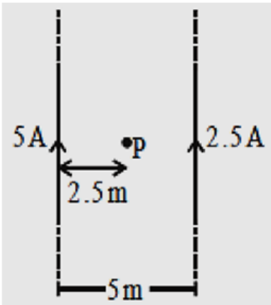
(1) 10^{-1} T

(2) 10^{-2} T

(3) 10^{2} T

(4) 10^{-3} T

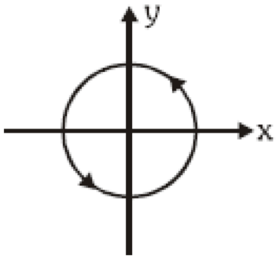
17. For adjoining figure. The magnetic field at point 'P' will be:-



(1) $\frac{\mu_0}{4\pi}\odot$ (2) $\frac{\mu_0}{\pi}\otimes$

(3) $\frac{\mu_0}{2\pi}\otimes$ (4) $\frac{\mu_0}{2\pi}\odot$

18. Current i = 2.5 A flows along the circle $x^2 + y^2 = 9\ cm^2$ (here x & y in cm) as shown



Magnetic field at point (0, 0, 4 cm) is :-

(1) $(36\pi \times 10^{-7}T)\hat{k}$

(2) $(36\pi \times 10^{-7}T)(-\hat{k})$

(3) $\left(\frac{9\pi}{5} \times 10^{-7}T\right)\hat{k}$

(4) $\left(\frac{9\pi}{5} \times 10^{-7}T\right)(-\hat{k})$

19. A mass M is lowered with the help of a string by a distance h at a constant acceleration g/2. The work done by the string will be :-

(1) $\frac{Mgh}{2}$

(2) $-\frac{Mgh}{2}$

(3) $\frac{3Mgh}{2}$

(4) $-\frac{3Mgh}{2}$

20. If K.E. increases by 3%. Then momentum will increase by :-

(1) 1.5% (2) 9%

(3) 3% (4) 2%

21. A uniform chain has a mass m and length l. It is held on a frictionless table with $\left(\frac{1}{6}\right)$ of its length hanging over the edge. The work done in just pulling the hanging part back on the table is:-

(1) $\frac{mgl}{72}$ (2) $\frac{mgl}{36}$ (3) $\frac{mgl}{12}$ (4) $\frac{mgl}{6}$

22. A 50 kg man with 20 kg load on his head climbs up 20 steps of 0.25 m height each. The minimum work done by the man on the block during climbing is :- ($g = 10\ m/s^2$)

(1) Zero (2) 350 J

(3) 1000 J (4) 3540 J

23. A particle moves under the effect of a force F = 8x from x = 0 to x = 5. The work done in the process is :

(1) 50 J (2) 25 J

(3) 100 J (4) 75 J

24. When a spring is stretched by 2cm, it stores 100 J of energy. If it is stretched further by 2 cm more, the stored energy will be increased by :-

(1) 100 J (2) 200 J

(3) 300 J (4) 400 J

25. A beaker is completely filled with water at 4°C. It will overflow :-

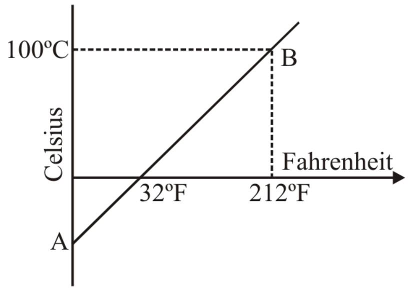
(1) When heated but not when cooled

(2) When cooled but not when heated

(3) Both when heated or cooled

(4) Neither when heated nor when cooled

26. The graph AB shown in figure is a plot of temperature of a body in degree celsius and degree Fahrenheit. The slope of line AB is :-



(1) $\frac{9}{5}$ (2) $\frac{5}{9}$ (3) $\frac{1}{9}$ (4) $\frac{3}{9}$

27. On a new scale of temperature (which is linear) and called the W scale, the freezing and boiling points of water are 39°W and 239°W respectively. What will be the temperature on the new scale, corresponding to a temperature of 39°C on the Celsius scale :-

(1) 200°W

(2) 139°W

(3) 78°W

(4) 117°W

28. On centigrade scale the temperature of a body increases by 40°. The increase in temperature on fahrenheit scale is :

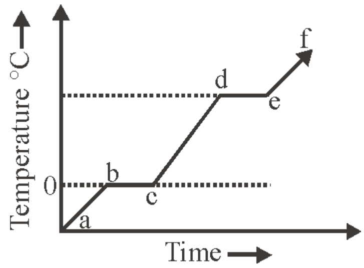
(1) 100°

(2) 80°

(3) 60°

(4) 72°

29. The following figure represents the temperature versus time plot for a given amount of a substance when heat energy is supplied to it at a fixed rate and at a constant pressure . Which part of the above plot represent a phase change :-



(1) a to b and e to f (2) b to c and c to d

(3) d to e and e to f (4) b to c and d to e

30. Calculate the amount of heat (in calories) required to convert 5 gm of ice at 0°C to steam at 100°C.

(1) 3100 (2) 3200 (3) 3600 (4) 4200

31. A drilling machine of power P watt is used to drill a hole in copper block of mass M Kg if the specific heat of copper is 5 J $Kg^{-1}°C^{-1}$ and 40% of the energy is lost due to heating of the machine, the rise in the temperature of the block in T seconds will be (in °C) (assume the rest energy goes in heating block) :-

(1) $\frac{0.6PT}{Ms}$ (2) $\frac{0.6P}{MsT}$

(3) $\frac{0.4PT}{Ms}$ (4) $\frac{0.4P}{MsT}$

32. If latent heat of a material is 40 $\frac{cal}{gm}$ then in SI unit :-

(1) 167200 (2) 1250

(3) 9.6 (4) 1780

33. SI unit of specific heat is :

(1) $\frac{Joule}{gmK}$ (2) $\frac{cal}{gmK}$

(3) $\frac{Kcal}{KgK}$ (4) $\frac{Joule}{KgK}$

34. If 500 cal heat is supply to 10 gm water at 60°C what will be final temp :-

(1) 110°C (2) 100°C

(3) 50°C (4) 40°C

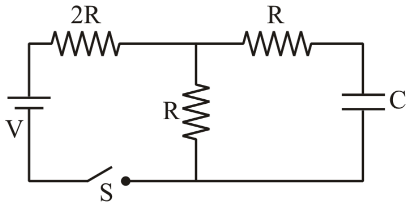
35. If water equivalent of a container is 10 gm then find heat required to change temp of it by 10°C :-

(1) 100 Joule (2) 100 Cal

(3) 150 Cal (4) 150 J

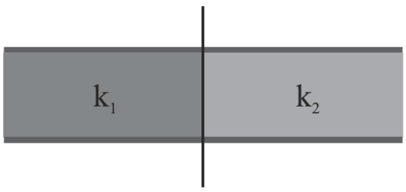
SECTION - B (PHYSICS)

36. Current drawn from the battery just after switch 'S' being closed :-



(1) $\frac{2V}{5R}$ (2) $\frac{3V}{5R}$ (3) $\frac{4V}{5R}$ (4) $\frac{6V}{4R}$

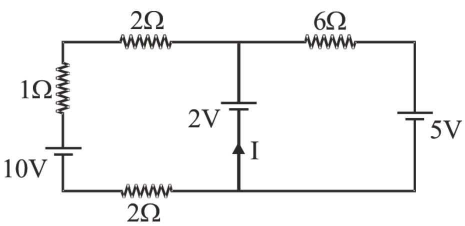
37. A parallel plate capacitor with air as medium between the plates has a capacitance of 10µF. The area of capacitor is divided into two equal halves and filled with two media as shown in the figure having dielectric constant $k_1 = 2$ and $k_2 = 4$. The capacitance of the system will now be :-



(1) 10 µF (2) 20 µF

(3) 30 µF (4) 40 µF

38.

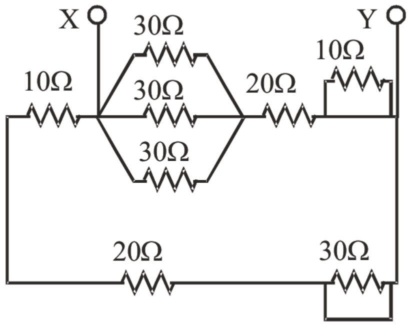


Find the value of I ?

(1) −1.4 A (2) −2.1 A

(3) 2 A (4) −4.2 A

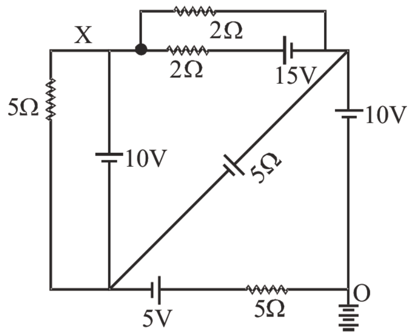
39. Net resistance between X and Y is –



(1) 5 Ω (2) 10 Ω

(3) 15 Ω (4) 60 Ω

40. In given circuit if point O is earthed, the potential of point x is :-



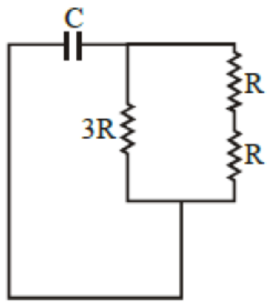
(1) 10 V

(2) 15 V

(3) 22 V

(4) 12.5 V

41. The time constant of the given circuit is :



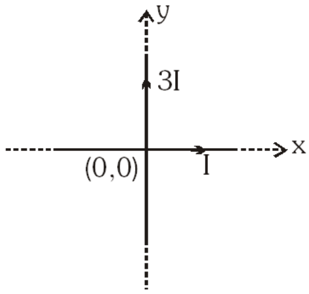
(1) $\frac{3RC}{5}$

(2) $\frac{6RC}{5}$

(3) $\frac{5RC}{6}$

(4) None of the above

42. The equation of line on which magnetic field is zero due to system of two perpendicular infinitely long current carrying straight wires, is :-



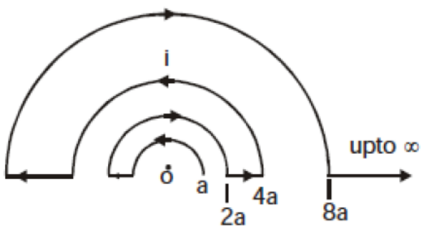
(1) $x = y$ (2) $x = 2y$

(3) $x = 3y$ (4) $3x = y$

43. There are 8 drops of a conducting fluid. Each has radius r and they are charged to potential 1 volt. They are then combined to form a bigger drop. Find potential of big drop.

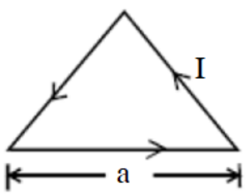
(1) 1 V (2) 4 V (3) 2 V (4) 8 V

44. A conducting wire is bent in the form of concentric semicircles as shown in the figure. The magnetic field at the point O is :-



(1) Zero (2) $\mu_0 i/6a$ (3) $\mu_0 i/a$ (4) $\mu_0 i/4a$

45. An equilateral triangle of side 'a carries a current 'I'. Magnetic field at point P which is vertex of triangle :-



(1) $\dfrac{\mu_0 I}{2\sqrt{3}\pi a} \odot$ (2) $\dfrac{\mu_0 I}{2\sqrt{3}\pi a} \otimes$

(3) $\dfrac{9}{2}\left(\dfrac{\mu_0 I}{\pi a}\right) \odot$ (4) $\dfrac{9}{2}\left(\dfrac{\mu_0 I}{\pi a}\right) \otimes$

46. A position-dependent force $F = 7 - 2x + 3x^2$ newton acts on a small body of mass 2 kg and displaces it from $x = 0$ to $x = 5$ m. The work done in joule is

(1) 70 (2) 270

(3) 35 (4) 135

47. An electric motor creates a tension of 4500 N in hoisting cable & reels it at a rate of 2m/s. What is the power of electric motor?

(1) 9 W

(2) 9 kW

(3) 225 W

(4) 9000 H.P

48. 10 gm of Ice at 0°C is mixed with 30 gm of water at 40°C. The final temperature of the mixture is :

(1) 8°C (2) 10°C

(3) 15°C (4) 20°C

49. Two identical bodies are made of a material for which the heat capacity increases with temperature. One of these is at 80°C, while the other one is at 0°C. If the two bodies are brought into contact, then, assuming no heat loss, the final common temperature is :-

(1) Less than 40°C but greater than 0°C

(2) 0°C

(3) 40°C

(4) More than 40°C

50. If specific heat of a substance is infinite, it means :-

(1) heat is given out

(2) heat is taken in

(3) No change in temp is possible

(4) temp will increase

Topic : GOC-II (CURRENT), HYDROCARBON, HALOGEN AND OXYGEN COMPOUNDS, PERIODIC TABLE, CHEMICAL BONDING (UPTO VSEPR THEORY), [BACK UNIT] : ELECTROCHEMISTRY

SECTION-A (CHEMISTRY)

51. The nomenclature of ICl is iodine chloride because

(1) Size of I < Size of Cl

(2) Atomic number of I > Atomic number of Cl

(3) E.N. of I < E.N. of Cl

(4) E. A. of I < E. A. of Cl

52. If IE_1 of Na is +5.1 eV, then ΔHe_g of Na^+ would be

(1) Less than 5.1 eV

(2) +5.1 eV

(3) –5.1 eV

(4) More than 5.1 eV

53. Which of the following process can be endothermic.

(I) $O + e^- \longrightarrow O^-$

(II) $O^- + e^- \longrightarrow O^{-2}$

(III) $Ne + e^- \longrightarrow Ne^-$

(IV) $N + e^- \longrightarrow N^-$

(1) I, II (2) II, III

(3) II, III, IV (4) I, II, III, IV

54. Which set of elements shows positive electron gain enthalpy

(1) He, N, O (2) Ne, N, Cl

(3) O, Cl, F (4) N, He, Ne

55. Valency of which of the following element is not equal to the valence electron

(1) Na (2) Mg

(3) C (4) O

56. Which among the following pair does not follows octet rule :

(1) CCl_4, CO_2 (2) IBr, SF_2

(3) BCl_3, PCl_5 (4) NaCl, H_2O

57. **Assertion :** The first ionisation energy of aluminum is lower than that of Mg.

Reason : Atomic radius of Al is larger than that of Mg.

(1) Assertion is correct, reason is incorrect

(2) Assertion is incorrect, reason is correct

(3) Both are correct and reason is correct explanation

(4) Both are correct and reason is not correct explanation

58. Among O, S, F, Cl most and least negative ΔH_{eg} will be respectively of :-

(1) Cl, S (2) F, S (3) Cl, O (4) F, O

59. Which application of EN is incorrect ?

(1) Acidic character of oxide/hydroxide increases left to right in period

(2) Acidic character of hydride of non metal increases down side in group

(3) Acidic character of metal hydride increases downside in group

(4) Basic character of oxide/hydroxide increases downside in group

60. Correct order of electronegativity of N, O, P, S elements.

(1) O < N < S < P

(2) O > N > S > P

(3) O > N > P > S

(4) O > P > N > S

61. What is oxidation state and covalency of Al in $[Al\,Cl(H_2O)_5]^{2+}$ is :-

(1) +3, 3 (2) +3, 6

(3) +2, 6 (4) +2, 5

62. Incorrect statement for covalent bond by VBT

(1) Half filled orbitals participate in overlapping

(2) Covalent bond is non-directional bond

(3) Strength of bond depend on extent of overlapping

(4) Axial overlapping is stronger than lateral overlapping

63. The best conductor of electricity is a 0.1 M solution of :-

(1) H_2SO_4

(2) CH_3COOH

(3) CH_3CH_2COOH

(4) Boric acid

64. Which has maximum conductance :-

(1) $[Co(NH_3)_6]Cl_3$

(2) $[Co(NH_3)_5Cl]Cl_2$

(3) $[Co(NH_3)_4Cl_2]Cl$

(4) None of these

65. An aqueous solution of Na_2SO_4 is electrolysed using Pt electrodes. The products at the cathode and anode are respectively :-

(1) H_2, SO_2

(2) O_2, NaOH

(3) H_2, O_2

(4) O_2, SO_2

66. The standard electrode potential for the reactions

$Ag^+_{(aq)} + e^- \longrightarrow Ag_{(s)}$ and $Sn^{2+}_{(aq)} + 2e^- \longrightarrow Sn_{(s)}$

at 25°C are 0.80 volt and – 0.14 volt respectively then EMF of the cell $Sn \mid Sn^{2+}(1M) \parallel Ag^+(1M) \mid Ag$ is :

(1) 0.66 volt

(2) 0.80 volt

(3) 1.08 volt

(4) 0.94 volt

67. The resistance of 0.5 M solution of an electrolyte in a cell was found to be 50 Ω. If the electrodes in the cell are 2.2 cm apart and have an area of 4.4 cm^2 then the molar conductivity (in $S\ m^2\ mol^{-1}$) of the solution is

(1) 0.2

(2) 0.02

(3) 0.002

(4) None of these

68. The moles of electron required to deposit 1 g equivalent of aluminum (atomic wt. = 27) from a solution of aluminum chloride will be :

(1) 3

(2) 1

(3) 4

(4) 2

69. Which of the following reaction is not feasible ?

(1) $CuSO_4 + Fe \longrightarrow FeSO_4 + Cu$

(2) $CuSO_4 + Zn \longrightarrow ZnSO_4 + Cu$

(3) $CuSO_4 + 2Ag \longrightarrow Ag_2SO_4 + Cu$

(4) $CuSO_4 + Mg \longrightarrow MgSO_4 + Cu$

70. Amount of charge that can deposit 108 g of silver from $AgNO_3$ solution is :

(1) 1 Ampere

(2) 1 Coulomb

(3) 1 Faraday

(4) None of these

71. $\lambda_m^\infty (NH_4OH)$ is equal to :

(1) $\lambda_m^\infty (NH_4Cl) + \lambda_m^\infty (NaOH) - \lambda_m^\infty (NaCl)$

(2) $\lambda_m^\infty (NH_4Cl) + \lambda_m^\infty (NaCl) - \lambda_m^\infty (NaOH)$

(3) $\lambda_m^\infty (NaOH) + \lambda_m^\infty (NaCl) - \lambda_m^\infty (NH_4Cl)$

(4) None of these

72. Equivalent conductivity (λ_{eq}) of $Fe_2(SO_4)_3$ is related to molar conductivity (λ_m) by the expression :

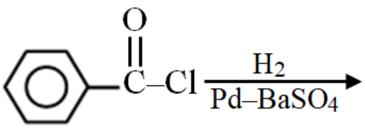
(1) $\lambda_{eq} = \lambda_m$

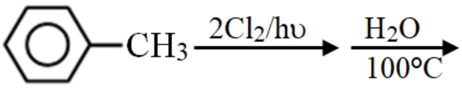
(2) $\lambda_{eq} = \frac{\lambda_m}{3}$

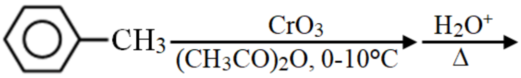
(3) $\lambda_{eq} = 6\lambda_m$

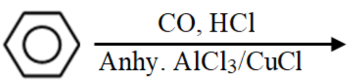
(4) $\lambda_{eq} = \frac{\lambda_m}{6}$

73. Which of the following can be used to prepare benzaldehyde :

(I) 

(II) 

(III) 

(IV) 

(1) I, II (2) II, III, IV

(3) I, III, IV (4) I, II, III, IV

74. For the reaction :-

$$Ph{-}CHO + Ph{-}\underset{\underset{O}{\|}}{C}{-}CH_3 \xrightarrow[293K]{\overset{\ominus}{O}H} \text{Major}$$

The major product is :

(1) $Ph{-}\overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}}{-}CH_2{-}\underset{\underset{O}{\|}}{C}{-}Ph$

(2) $Ph{-}\underset{\underset{O}{\|}}{C}{-}O^{\ominus} + Ph{-}CH_2{-}OH$

(3) $Ph{-}\underset{\underset{OH}{|}}{CH}{-}\overset{\overset{O}{\|}}{C}{-}Ph$

(4) $Ph{-}CH{=}CH{-}\overset{\overset{O}{\|}}{C}{-}Ph$

75. Which of the following reagents is/are used in the given reaction ?

$RCHO \longrightarrow RCOOH$

(1) $KMnO_4/H^+$

(2) Potassium dichromate

(3) Tollen's reagent

(4) All of the above

76. Which of the following statements is/are correct?

(1) Aldehydes are generally oxidised under vigorous conditions

(2) Ketones are easily oxidised to carboxylic acids even under mild oxidising agents

(3) Oxidation of ketones involves carbon-carbon bond cleavage to give a mixture of carboxylic acids having lesser number of c-atoms than the parent ketones.

(4) All of the above

77. Which of the following compound gives yellow ppt with I_2 + NaOH :

(1) $CH_3 - CH_2 - OH$

(2) $CH_3 - \overset{\overset{OH}{|}}{CH} - CH_3$

(3) $CH_3 - \overset{\overset{Cl}{|}}{CH} - CH_3$

(4) All of these

78. Which is correct reaction for major product :

(a) $CH_3{-}CH_2{-}\overset{\overset{F}{|}}{CH}{-}CH_3 \xrightarrow[\Delta]{C_2H_5O^-} CH_3{-}CH_2{-}CH{=}CH_2$

(b) $CH_3{-}CH_2{-}\underset{\underset{F}{|}}{CH}{-}CH_3 \xrightarrow[\Delta]{C_2H_5O^-} CH_3{-}CH{=}CH{-}CH_3$

(c) $CH_3{-}\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}{-}CH_2OH + \text{Conc.}H_2SO_4 \xrightarrow{\Delta} CH_3{-}\underset{\underset{CH_3}{|}}{C}{=}CH{-}CH_3$

(1) a, b, c (2) a, c (3) b, c (4) a, b

79. Which of the following reactions of alkanols does not involve C–O bond breaking

(1) $CH_3CH_2OH + SOCl_2$

(2) $CH_3CH(OH)CH_3 + PBr_3$

(3) $CH_3CH_2OH + CH_3COOH$

(4) $ROH + HX$

80. Which of the following statement is incorrect?

(1) Acetaldehyde & acetone can be differentiated by tollen's test.

(2) Primary and tertiary Alcohol can be differentiated by dichromate test.

(3) Acetic acid and ethanol can be differentiated by $NaHCO_3$.

(4) Acetaldehyde and formic acid can be differentiated by Tollen's test.

81. **Statement-1 :** Aryl halides are less reactive than alkyl halide for nucleophilic substitution reaction.

Statement-2 : In alkyl halides, the C-X bond acquires a partial double bond character due to resonance.

(1) Statement-1 is incorrect, statement-2 is correct.

(2) Statement-1 is correct, statement-2 is incorrect.

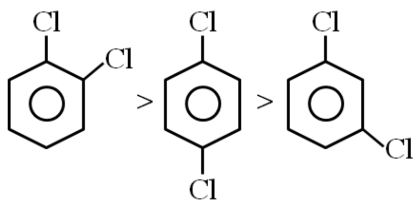
(3) Both statements are correct.

(4) Both statements are incorrect.

82. Correct order of boiling point.

(1) $CH_3–Cl > CH_3–CH_2–Cl > CH_3–CH_2–CH_2–Cl$

(2) $CH_3–CH_2–CH_2–CH_2–Br < CH_3–C(CH_3)(Br)–CH_3$

(3) 

(4) None of the above

83. The purity of $CHCl_3$ can be checked by :-

(1) Treating $CHCl_3$ by NaOH

(2) Treating $CHCl_3$ by HCl

(3) Treating $CHCl_3$ with aq. $AgNO_3$

(4) Treating $CHCl_3$ by C_2H_5OH

84. Which of the following will form alkenes as a major product?

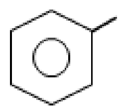
(1) $CH_3–CH_2–Br + CH_3–CH(CH_3)–ONa$

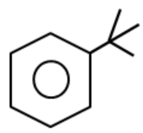
(2) $CH_3–Cl + CH_3–C(CH_3)_2–ONa$

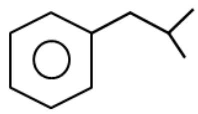
(3) $CH_3–CH_2–Cl + CH_3ONa$

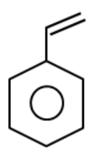
(4) $CH_3–C(CH_3)_2–Cl + CH_3ONa$

85. Which of the following will not give benzoic acid on reaction with $KMnO_4/H^+$?

(1) 

(2) 

(3) 

(4) 

SECTION-B (CHEMISTRY)

86. Which of the following order is Incorrect :

(1) HOF > HOCl > HOBr > HOI Acidic nature

(2) LiOH < NaOH < KOH < RbOH < CsOH Basic nature

(3) KH < RbH < CsH Basic nature

(4) $H_2O > H_2S > H_2Se > H_2Te$ Acidic nature

87. Maximum excited state can be shown by Xenon ?

(1) 1^{st} E.S.

(2) 2^{nd} E.S.

(3) 4^{th} E.S.

(4) It has only ground state

88. Which is incorrect statement :

(1) He belongs to s-block but its position in p-block.

(2) All inert gas have ns^2np^6, fully filled electronic configuration.

(3) Electronic configuration of Gd_{64} is $4f^7 5d^1 6s^2$

(4) Oxygen have least negative ΔH_{eg} in its group

89. Incorrect statement about Be and B ionisation energy ?

(1) For different principal quantum level s-electron is more attracted then p-electron

(2) Penetration effect of s-electron of Be is higher than p-electron of B

(3) 2p-electron of boron is very less shielded by inner electrons

(4) 1 and 3 both

90. IE_1 values of 3rd period elements Na, Mg and Si are respectively 496, 737 and 786 kJ/mole then IE_1 of Al is in between :-

(1) 496 kJ - 737 kJ

(2) 737 kJ - 786 kJ

(3) Above 786 kJ

(4) Below 496 kJ

91. Standard reduction electrode potentials of three metals A, B and C are +0.5 V, –3.0 V and –1.2 V respecitvely. The reducing power of these metals are :

(1) B > C > A

(2) A > B > C

(3) C > B > A

(4) A > C > B

92. Electrolytic conduction differs from metallic conduction from the fact that in the former

(1) The resistant increases with increasing temperature

(2) The resistance decreases with increasing temperature

(3) The resistance remains constant with increasing temperature

(4) The resistance is independent of the length of the conductor

93. If the same charge is passed through the solution of Fe^{+2} and Cu^{+2}, then 5.6 g Fe is deposited. Find weight of Cu deposited? (Given : Atomic mass of Cu = 63.5 and Fe = 56 g)

(1) 6.35 g

(2) 63.5 g

(3) 3.175 g

(4) 31.75 g

94. Each of the three metals x, y and z were put in turn into aqueous solution of the other two.

x + salt of y (or z) $\longrightarrow$ y (or z) + salt of x.

Which one of the following observation is incorrect?

(1) y + salt of x $\longrightarrow$ no action observed

(2) y + salt of z $\longrightarrow$ z + salt of y

(3) z + salt of x $\longrightarrow$ x + salt of z

(4) z + salt of y $\longrightarrow$ no action observed

95. When a lead storage battery is discharged.

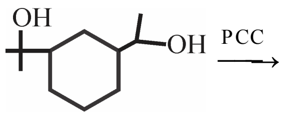
(1) SO_2 is evolved

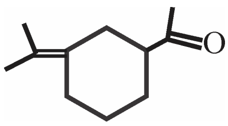
(2) Lead sulphate is consumed

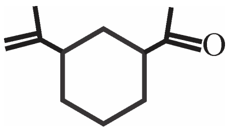
(3) Lead is formed

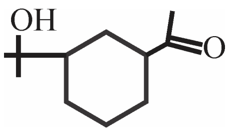
(4) Sulphuric acid is consumed

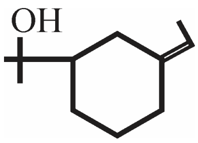
96. What is product of the following reaction?

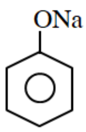


(1) 

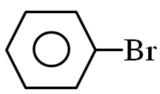
(2) 

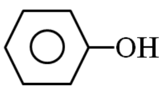
(3) 

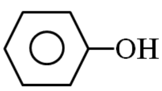
(4) 

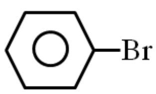
97. CH_3-CH_2-Br +  $\longrightarrow A \xrightarrow[\text{HBr}]{\text{cold \& conc.}} B \ \& \ C$

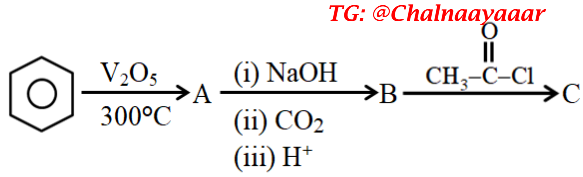
B and C are :-

(1) CH_3-CH_2-Br + 

(2) CH_3-CH_2-OH + 

(3) CH_3-CH_2-Br + 

(4) CH_3-CH_2-OH + 

98. 

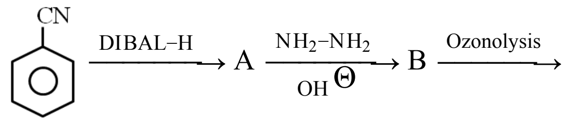
Incorrect statement ?

(1) A reacts with CH_3MgBr to form methane.

(2) B reacts with PhOH to form SALOL.

(3) A reacts with PCl_5 to form tri phenyl phosphate.

(4) C is an antiseptic.

99. 

products

Incorrect statement?

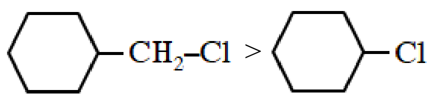
(1) A is benzyl amine.

(2) End products contain glyoxal and methyl glyoxal.

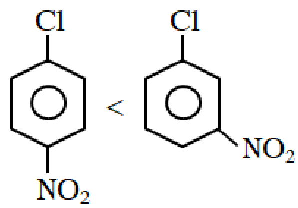
(3) B is oxidised by $KMnO_4/H^+$ to give benzoic acid.

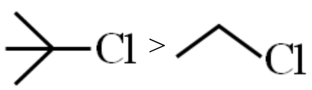
(4) B reacts with chromyl chloride followed by hydrolysis to form benzaldehyde.

100. Which of the following order of rate of nucleophilic substitution is incorrect?

(1)  (S_N2)

(2) $CH_3-CH_2-Cl < Ph-CH_2-Cl$ (S_N2)

(3)  (S_NAr)

(4)  (S_N1)

Topic : MBI (GENETIC CODE TO COMPLETE), BIOTECH PRINCIPLES (COMPLETE), BIODIVERSITY AND CONSERVATION, ENVIRONMENTAL ISSUE, CELL BIOLOGY, MICROBES IN HUMAN WELFARE, [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-I)

101. Given below are three statements (A-C) each with one of more blanks. Select the which correctly fills the blanks in the statements :-

(A) ______a traditional drink of some parts of South India is made by fermenting sap from palms

(B) Citric acid is obtained through the fermentation carried out by _____

(C) In______the fungus forms a mantle on the surface of the roots

(1) A–Toddy B–Aspergillus niger, C–Ectomycorrhiza

(2) A–Wine B–Aspergillus niger, C–Endomycorhiza

(3) A–Beer B–Aspergillus niger, C–Endomycorhize

(4) A–Rum B–Aspergillus niger, C–Endomycorhiza

102. Big holes in Swiss cheese are made by a :-

(1) Virus

(2) Bacteria that produce methane gas

(3) Bacteria produce large amount of CO_2.

(4) Fungus that releases lot of gases during its metabolic activities.

103. Free living fungus trichoderma can be used for :-

(1) Killing insects

(2) Biological control of plant disease.

(3) Controlling butterfly caterpillars

(4) Producing antibiotics

104. Find the mismatch :-

(1)	Sequence Annotation	–	Blind approach of simply sequencing the whole set of genome
(2)	Non-repetitive DNA	–	Shed light on chromosome structure dynamics and evolution
(3)	Polymorphisms	–	Are inheritable and arises due to mutations
(4)	VNTR	–	Copy number varies from chromosomes to chromosome in an individual

105. **Assertion :** In the first phase of translation itself amino acids are activated in the presence of ATP

Reason : During translation for the formation of a peptide bond energy is required.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

106. Which of the following m-RNA is translated completely :-

(A) 5' AUG UGA UUA AAG AAA 3'

(B) 5' AUG AUA UUG CCC UGA 3'

(C) 5' AGU UCC AGA CUC UAA 3'

(D) 5' AUG UAC AGU AAC UAG 3'

(1) (A) and (B) (2) (B) and (D)

(3) (C) and (D) (4) (A) and (D)

107. How many ATP and GTP required for polymerization of 100 amino acids ?

(1) 100 ATP and 200 GTP

(2) 100 ATP and 100 GTP

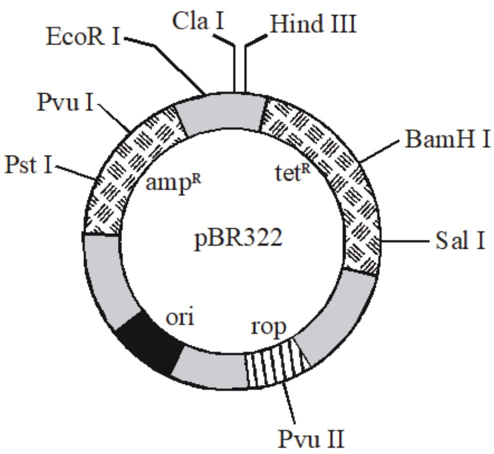
(3) 50 ATP and 200 GTP

(4) 200 ATP and 400 GTP

108. Approximately how many restriction enzymes have been isolated from different (over 230) bacteria ?

(1) 300 (2) 600 (3) 750 (4) 900

109. The figure below is the diagrammtic representation of the E.coli vector pBR 322. Which one of the given options correctly identifies its certain component(s) :-



(1) Ori-original restriction enzyme

(2) Rop-reduced osmotic pressure

(3) Hind III, EcoRI-selectable markers

(4) amp^R, tet^R – antibiotic resistance genes

110. In gel electrophoresis :-

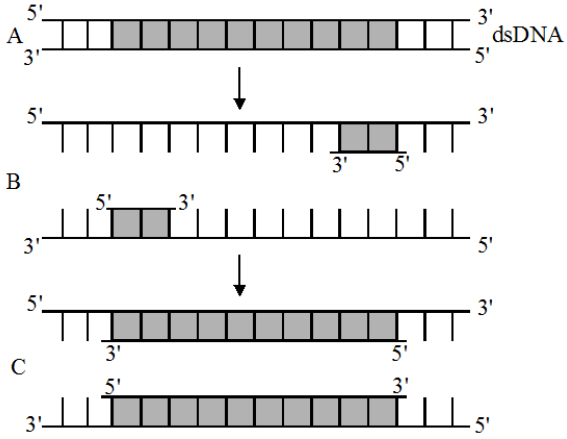
(1) DNA fragments are arranged according to length on gel by applying electric field

(2) DNA fragments are arranged according to charge

(3) DNA fragments are treated by alcoholic solution

(4) DNA fragments are destroyed

111. The figure below shows three steps (A,B, C) of Polymerase Chain Reaction (PCR). Select the option giving correct identification together with what it represents?



(1) C-Extension in the presence of heat stable DNA polymerase

(2) A-Annealing with two sets of primers

(3) B-Denaturation at a temperature of about 98°C separating the two DNA strands

(4) A-Denaturation at a temperature of about 50°C

112. I.... A... is the ability of a cell to take up foreign DNA.

II. The cell is treated with specific concentration of a divalent cation such as ...B... to increase efficiency.

III. In... C... method recombinant DNA is directly injected into the nucleus of an animal cell.

The most appropriate option regarding A, B and C is

(1)	A-Competency	B-Calcium	C-gene gun method
(2)	A-Transformation	B-Sodium	C-microinjection method
(3)	A-Competency	B-Calcium	C-microinjection method
(4)	A-Transformation	B-Sodium	C-gene gun method

113. UCG codes for :-
(1) Valine (2) Serine
(3) Glutamic acid (4) Leucine

114. A classical example of point mutation is :-
(1) Cridochat syndrome
(2) Down syndrome
(3) Leukemia
(4) Sickle cell anemia

115. In eukaryotes, the regulation could be exerted at which of the following levels?
(A) Replication level
(B) transcriptional level
(C) processing level (regulation of splicing)
(D) transport of mRNA from nucleus to the cytoplasm
(E) translational level
(1) A, B, C, D & E (2) A, B, D & E
(3) B, C, D & E (4) B, C & D

116. Product of y-gene in lac operon increases permeability of cell to -
(1) β-galactosidase (2) Glucose
(3) Transacetylase (4) β-galactosides

117. HGP was started in the year :-
(1) 1900 (2) 1990 (3) 2003 (4) 2006

118. How many genes are on chromosome-1 ?
(1) 2968 (2) 3000
(3) 231 (4) 3164

119. Polymorphism in DNA sequence is the basis of _____ of human genome as well as of _____.
(1) Genetic mapping, DNA fingerprinting
(2) Protein synthesis, DNA fingerprinting
(3) Genetic mapping, RNA synthesis
(4) Replication, Transcription

TG: @Chalnaayaaar

120. If instead of 99.9%, only 90% of base sequences among humans were same then how many base sequences would be different in genome ?
(1) 3×10^5 (2) 3×10^6
(3) 3×10^7 (4) 3×10^8

121. The integration of how many of the following for products and services is called biotechnology?
natural science, organisms, cells, parts of cells, molecular analogues
(1) Five (2) Four
(3) Three (4) Two

122. First discovered Restriction endonuclease was ?
(1) Hind-II (2) Hind-I
(3) EcoR-1 (4) EcoR-2

123. **Statement-1 :** Each restriction endonuclease functions by 'inspecting' the length of a DNA sequence.
Statement-2 : You can see bright blue coloured bands of DNA in an ethidium bromide stained gel exposed to UV light.
(1) Both statement-1 and statement-2 are correct
(2) Both statement-1 and statement-2 are incorrect
(3) Statement-1 is correct and statement-2 is incorrect
(4) Statement-1 is incorrect and statement-2 is correct

124. How much volume of culture can be processed in bioreactors?
(1) 10-100 L
(2) 100-1000 ml
(3) 1000-10,000 ml
(4) 100-1000 L

125. Restriction endonuclease is obtained from :-

(1) Plants

(2) Fungi

(3) Bacteria

(4) Animals

126. How many primary follicles are left in each ovary at puberty ?

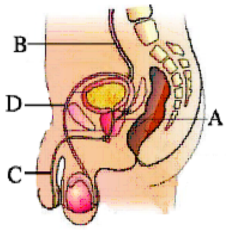
(1) 60,000 – 80, 000

(2) 4-lac

(3) 7-million

(4) 1,20,000

127. It is a diagrammatic sectional view of male reproductive system, In which identify common duct which forms by the fusion of duct of seminal vesicle and vasdeferens :



(1) A (2) B

(3) D (4) C

128. Androgens stimulate______.

(1) Spermiation

(2) Insemination

(3) Spermatogenesis

(4) Oogenesis

129. Implantation refers to :-

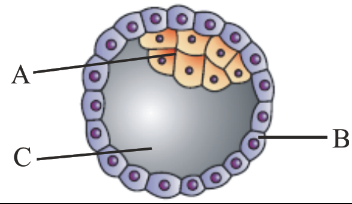
(1) embedement of blastocyst in endometrium

(2) embedement of blastocyst in myometrium

(3) embedement of blastocyst in perimetrium

(4) embedement of morula in cervical canal

130. In the given figure identify the parts labelled as A,B & C, and choose the correct option :-



	A	B	C
(1)	Inner cell mass	Trophoblast	Blastocoel
(2)	Trophoblast	Inner cell mass	Blastocoel
(3)	Blastomere	Trophoblast	Inner cell mass
(4)	Trophoblast	Blastocoel	Inner cell mass

131. Match the following and choose the correct answer :-

(A)	Pregnancy Hormone	(i)	Estrogen
(B)	Proliferation of Endometrium	(ii)	Ovary
(C)	Oogenesis	(iii)	Progesterone
(D)	Cleavage	(iv)	Forms blastomere

(1) A-(iv), B-(i), C-(iii), D-(ii)

(2) A-(iii), B-(ii), C-(iv), D-(i)

(3) A-(iii), B-(i), C-(ii), D-(iv)

(4) A-(iii), B-(iv), C-(ii), D-(i)

132. Mark the incorrect statement -

(A) Sertoli cells provide nourishment to dead sperm cells.

(B) Vagina secrete the alkaline fluid, to make easy passage for sperms.

(C) Fertilisation occure in fundus of female uterus.

(D) Oxytocin is the main parturation hormone.

(1) A, C (2) A, B, C

(3) B, C, D (4) A, D

133. Identify the correct matching pairs from the following.

i	Menstruation	Breakdown of endometrium
ii	Proliferative phase	Degeneration of endometrium and follicular development
iii	Secretory phase	Development of corpus luteum and increases secretion of progesterone
iv	Ovulatory phase	LH surge induces ovulation

(1) i and ii only

(2) ii and iii only

(3) iii and iv only

(4) i, iii and iv

134. The first movements of the fetus and the appearance of hair on the head are usually observed during:

(1) 2^{nd} month

(2) 3^{rd} month

(3) 5^{th} month

(4) 6^{th} month

135. Birth canal is formed by

(1) Cervical canal + Uterus

(2) Cervical canal + Vagina

(3) Cervical canal + Isthmus

(4) Cervical canal + Fallopian tube

SECTION-B (BIOLOGY-I)

136. Match the following list of organism and their importance

Column I		Column II	
(a)	streptokinase	(i)	Aphid
(b)	Monascus purpureus	(ii)	Ripening of Swiss cheese
(c)	Lady bird	(iii)	Clot buster
(d)	Propionibacterium sharmanii	(iv)	Production of blood - cholesterol lowering agents

(1) a-iv, b-iii c-ii, d-i

(2) a-iv, b-ii, c-i, d-iii

(3) a-iii, b-i, c-iv, d-ii

(4) a-iii, b-iv, c-i, d-ii

137. The last human chromosome to be sequenced was chromosome _______ in May _______

(1) 1, 2006

(2) 5, 2006

(3) 1, 2003

(4) Y, 2006

138. **Assertion :** A very low level of expression of lac operon has to be present in the cell all the time.

Reason : A very low level of expression of lac operon is required in a cell otherwise lactose cannot enter the cells.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

139. **Assertion :** The size of VNTR varies from .1 to 20 kb.

Reason : The number of tandem repeat show very high degree of polymorphism.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

140. EcoR I cuts the DNA between the same two bases on the opposite strands when following sequence is present in the DNA ?

(1) 5' G↓AATTG 3'
3' CTTAA↑C 5'

(2) 5'G↓AATTC 3'
3'CTTAA↑G 5'

(3) 5' C↓CC GGG 3'
3' GGG CC↑C 5'

(4) 5' G↓GATCC 3'
3' CCTAG↑G 5'

141. Which one of the following characteristics is generally not prefered for a cloning vector ?

(1) An origin of replication

(2) An antibiotic resistance marker

(3) Multiple restriction sites for a single restriction enzyme

(4) A high copy number

142. Insertional inactivation is used for :-

(1) Selection of transformed cells

(2) Selection of non-transformed cells

(3) Selection of recombinants

(4) Selection of vector

143. Arrange the steps of rDNA technology in correct order:-

I. Extraction of the desired gene product.

II. Amplification of gene of interest.

III. Isolation of desired DNA fragment.

IV. Ligation of DNA fragment into vector.

V. Insertion of rDNA into host.

Correct order is :-

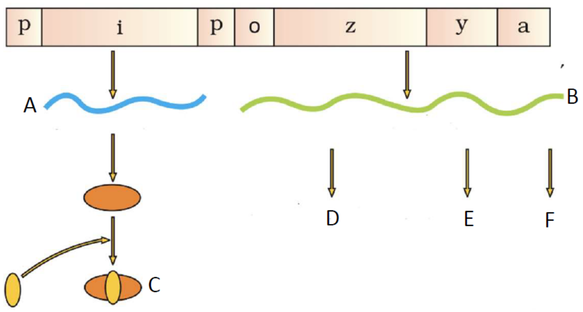
(1) I, II, III, IV, V

(2) V, IV, III, II, I

(3) III, II, IV, V, I

(4) III, IV, II, I, V

144.



Which of the following correctly identifies the structure A, B, C, D, E or F shown in above diagram?

(1) C = Active repressor protein,
D = β-galactosidase

(2) A = Lac mRNA, F = Transacetylase

(3) B = Polycistronic mRNA,
A = Repressor mRNA

(4) E = Transacetylase, B = lac mRNA

145. Which of the following step is optional in DNA fingerprinting?

(1) isolation of DNA

(2) digestion of DNA by restriction endonucleases

(3) increasing sensitivity by PCR

(4) separation of DNA fragments by electrophoresis

146. Urethra originates from ___(A)___ and passes through ___(B)___ to its external opening called urethral meatus. Fill 'A' and 'B' from following options:-

(1) (A) - Prostate, (B) - Seminal vesicles.

(2) (A) - Penis, (B) - Urinary bladder

(3) (A) - Urinary bladder, (B) - Penis

(4) (A) - Urinary bladder, (B) - Ureter

147. Consider the following statements and select the correct option.

Statement – A : Each fallopian tube extends from the periphery of each ovary to the uterus.

Statement – B : Ampullary region of the fallopian tube is narrower than that of infundibulum.

(1) Both statements – A and B are correct

(2) Both statements – A and B are incorrect

(3) Only statement – A is correct

(4) Only statement – B is correct

148. **Assertion (A) :** Presence or absence of hymen is not reliable indicator of virginity or sexual experience

Reason (R) : it can be broken down by a sudden fall or jolt , insertion of a vaginal tampon or active participation in some sports like horse riding

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is the not correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

149. Match the items in Column I with those in Column II and select the correct option given below :

Column I		**Column II**	
A.	Proliferative Phase	i.	Breakdown of endometrial lining
B.	Secretory Phase	ii.	Follicular Phase
C.	Menstruation	iii.	Luteal Phase

	A	B	C
(1)	ii	iii	i
(2)	i	iii	ii
(3)	iii	ii	i
(4)	iii	i	ii

150. Fertilization is fusion of–

(1) Diploid spermatozoan with diploid ovum to form diploid zygote.

(2) Haploid spermatozoan with diploid ovum to form diploid zygote.

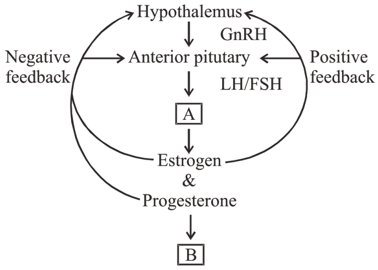
(3) Diploid spermatozoan with haploid ovum to form diploid zygote.

(4) Haploid spermatozoan with haploid ovum to form diploid zygote.

Topic : MBI (GENETIC CODE TO COMPLETE), BIOTECH PRINCIPLES (COMPLETE), BIODIVERSITY AND CONSERVATION, ENVIRONMENTAL ISSUE, CELL BIOLOGY, MICROBES IN HUMAN WELFARE, [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-II)

151. Find out 'A' and 'B' respectively.



(1) Uterus, ovary

(2) Ovary, Uterus

(3) Oviduct, Ovary

(4) Seminal vesicle, uterus

152. **Statement - I :** Placenta facilitates the supply of oxygen and nutrients to the mother and also removal of CO_2 and excretory waste of mother.
Statement - II : Placenta is functional unit between developing embryo and maternal body.
Choose the correct answer from the given options:

(1) Both statement - I and statement - II are correct.

(2) Statement - I is correct but statement II is not correct.

(3) Both statement - I and statement - II are not correct.

(4) Statement - I is not correct but statement - II is correct.

153. **Statement-I :** For normal fertility at least 40 percent of ejaculated sperm must have normal shape and size and at least 60 percent of them must show vigorous motility.
Statement-II : The semen along with the sperm is called seminal plasma.

(1) Both Statement-I and Statement-II are correct and Statement-II explanation Statement-I.

(2) Both Statement-I and Statement-II are correct and Statement-II does not explanation Statement-I.

(3) Statement-I is true and Statement-II is false.

(4) Both Statement-I and Statement-II is false.

154. The use of birth-control pills (oral contraceptives)

(1) reduces the incidence of ovulation.

(2) prevents fertilization by keeping sperm and egg physically separated by a mechanical barrier.

(3) promote implantation of an embryo.

(4) prevents sperm from exiting the male urethra.

155. All the following statement about ZIFT are correct but one is wrong find out that -

(1) It is zygote intra fallopian tube transfer

(2) Zygote is transferred in fallopian tube after IVF

(3) Early embryos upto 8 blastomeres can also be transferred in to fallopian tubes.

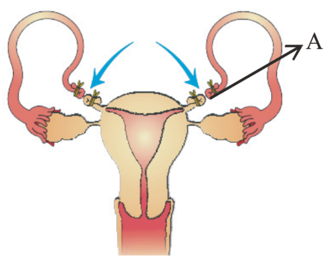
(4) Embryo with more than 8 blastomers are also transferred in the fallopian tubes.

156. Copper releasing IUD is :-

(1) LNG-20 (2) Lippes loop

(3) Mala-D (4) Multiload-375

157.



In above diagram which is shown and identify 'A'

(1) Vasectomy, Vasdeferens

(2) Tubectomy, Fallopian tube

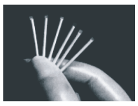
(3) Tubectomy, Ovary

(4) Vasectomy, Epididymis

158. given the figures below are related with contraceptive devices. Identify them and choose the correct option :-


(a)


(b)


(c)


(d)

	a	b	c	d
(1)	Condom for female	Condom for male	IUD	Implant
(2)	Condom for male	Condom for Female	Implant	IUD
(3)	Condom for female	Condom for male	Implant	IUD
(4)	Condom for male	Condom for Female	IUD	Implan

159. Match the column-I with column-II

Column-I		Column-II	
A	Oral pills	i	Block gametes transport
B	Hormonal Releasing IUDS	ii	Suppress sperm motility
C	Sterilisation methods	iii	Inhibit ovulation
D	Copper Releasing IUDS	iv	Make uterus unsuitable for implantation

(1) A-iv, B-ii, C-i, D-iii

(2) A-iii, B-iv, C-i, D-ii

(3) A-iii, B-ii, C-iv, D-i

(4) A-i, B-iv, C-ii, D-iii

160. Match the following and give the answer for correct match :-

A	ZIFT	i	Embryo with More than 8 blastomere stage transfer into the uterus
B	IUT	ii	Sperm is directly injected into the cytoplasm of ovum.
C	GIFT	iii	Zygote or embryo upto 8 blastomeres can be transferred into the fallopian tube
D	ICSI	iv	Transfer of an ovum collected from the ovary into the fallopian tube of the recipient.

(1) A-i B-ii C-iii D-iv

(2) A-iv B-iii C-ii D-i

(3) A-iv B-iii C-i D-ii

(4) A-iii B-i C-iv D-ii

161. **Statement-I:-** Copper releasing I.U.D's suppresses sperm motility.

Statement-II:- Progestasert and LNG-20 are the example of hormone releasing IUD.

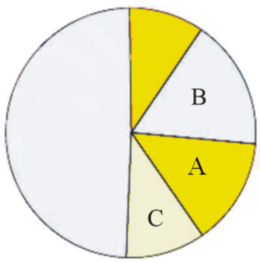
(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

162. In the following pie chart of global vertebrates diversity. What do A, B and C represent respectively?



(1) Birds, Fishes, Amphibians

(2) Mammals, Reptiles, Birds

(3) Reptiles, Birds, Amphibians

(4) Amphibians, Fishes, Reptiles

163. Match the column-I with column-II.

	Column-I		Column-II
A.	NEPA	i.	1986
B.	The air act	ii.	1992
C.	The water act	iii.	1981
D.	Earth summit	iv.	1974

(1) A-i, B-ii, C-iv, D-iii (2) A-i, B-iii, C-iv, D-ii

(3) A-ii, B-iv, C-iii, D-i (4) A-iii, B-ii, C-iv, D-i

164. Which of the following is not an exotic invading species in India ?

(1) *Lantana* (2) *Parthenium*

(3) *Eichhornia* (4) Makoi

165. Which of the followings is **not** a feature of biodiversity hot spot ?

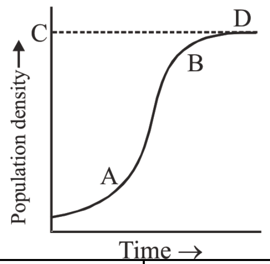
(1) High degree of endemism

(2) High species diversity

(3) Less area cover but higher impact on species conservation

(4) Low habitat loss

166. Which of the following is a correct match ?



	Column-I	Column-II
(i)	Carrying capacity	A
(ii)	Deceleration	B
(iii)	Asymptote	C
(iv)	Acceleration	D

Options :-

(1) (i)-B, (ii)-D, (iii)-C, (iv)-A

(2) (i)-B, (ii)-C, (iii)-D, (iv)-A

(3) (i)-D, (ii)-C, (iii)-A, (iv)-B

(4) (i)-C, (ii)-B, (iii)-D, (iv)-A

167. Which is not a character of population ?

(a) Birth (b) Stratification

(c) Mortality (d) Population density

(1) a and b

(2) b and c

(3) c and d

(4) a, c and d

168. The increase in concentration of the toxicant at successive trophic levels is referred to as:

(1) Eutrophication

(2) Bioremediation

(3) Biotransformation

(4) Biomagnification

169. The prime contaminants in lakes eutrophied by sewage and agricultural wastes are :-

(1) Sulphates and phosphates

(2) Nitrates and sulphates

(3) Nitrates and phosphates

(4) Nitrates and carbonates

170. Select the false statement regarding waste water treatment ?

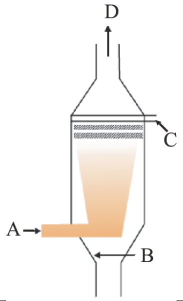
(1) Primary treatment basically involve physical removal of large & small particles.

(2) Flocs that are formed during secondary treatment possess mainly anaerobic bacteria and fungi

(3) During activated sludge treatment a small amount of activated sludge is pumped back into aeration tank

(4) The major part of activated sludge is pumped into anaerobic digesters where it produces biogas.

171. Given below is the diagrammatic sketch of certain type of scrubber. Identify the arrows labelled A,B,C and D and select the right option about them :-



	A	B	C	D
(1)	Particulate matter	Dirty Air	Water/lime spray	Clean Air
(2)	Lime spray	Particulate matter	Dirty Air	Clean Air
(3)	Dirty Air	Particulate matter	Lime spray	Clean Air
(4)	Dirty Air	Water/lime spray	Particulate matter	Clean Air

172. Which of the following is not a 'green house gas' ?
(a) Carbon dioxide (b) Oxygen
(c) Water vapour (d) Nitrogen oxide
(e) CFC
(1) a and d both (2) Only e
(3) c and e both (4) Only b

173. **Statement-I :-** In India, 1000 varieties of mango show species diversity.
Statement-II :- India is a megadiversity country.
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and Statement-II are correct.

174. Which of the following is not an advantage of CNG over diesel/petrol?
(1) It burns more efficiently.
(2) It is cheap
(3) Cannot be adulterated
(4) Easy to lay down pipelines for delivery.

175. Which of these is not a pollutant ?
(1) Thermal waste water (2) Noise
(3) Stratospheric ozone (4) Smog

176. In which phase, chromosome duplication occurs ?
(1) G_1-Phase (2) S-Phase
(3) G_2-Phase (4) M-Phase

177. **Assertion :-** Content of nucleolus remain continuous with the content of nucleoplasm.
Reason :- Nucleolus is a membrane-less structure.
(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

178. Which of the following is not included in 'Evil Quartet' ?
(1) Habitat loss and Fragmentation
(2) Over-exploitation
(3) Emissions of green house gases
(4) Co-extinction

179. Which of the following green house gas does not have natural source?
(1) CO_2 (2) CH_4 (3) N_2O (4) CFC

180. A stable ecosystem has :-
(a) High species diversity.
(b) High productivity.
(c) High variations in total biomass.
(d) Low resilience.
(e) High resistance.
(1) (a), (b), & (e) only
(2) (a), (c) & (e) only
(3) (a), (b), (c) & (d) only
(4) All except (d)

181. Which microbes play major role in decomposition of organic waste during sewage treatment ?
(1) Heterotrophic, aerobic
(2) Autotrophic, aerobic
(3) Heterotrophic, anaerobic
(4) Autotrophic, anaerobic

182. If there are 250 snails in a pond and within a year their number increases to 2500 by reproduction. What should be their birth per snail per year?
(1) 10 (2) 9 (3) 25 (4) 15

183. Thickness of ozone is measured in :
(1) Decibel unit
(2) Dobson unit
(3) Kg per m^2
(4) Atmospheric pressure at sea level

184. Match the column-I with column-II.

	Column-I		Column-II
(A)	Edward Wilson	(i)	Global biodiversity
(B)	Paul Ehrlich	(ii)	'Biodiversity' term
(C)	A.V. Humboldt	(iii)	'Rivet-Popper' hypothesis
(D)	Robert May	(iv)	Species-Area relationship

(1) (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)
(2) (A)-(ii), (B)-(i), (C)-(iii), (D)-(iv)
(3) (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)
(4) (A)-(i), (B)-(ii), (C)-(iv), (D)-(iii)

185. Match column-I with column-II and choose the correct option.

	Column-I (Chromosome)		Column-II (Position of Centromere)
(A)	Metacentric	(I)	At the tip
(B)	Submetacentric	(II)	Almost near the tip
(C)	Acrocentric	(III)	At the middle
(D)	Telocentric	(IV)	Slightly away from the middle

(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(3) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

SECTION-B (BIOLOGY-II)

186. **Statement-I :** The menstrual flow is result of breakdown of endometrial lining of the uterus and its blood vessels.
Statement-II : Menstrual flow lasts for 3-5 days.
(1) Statement I and II both are correct
(2) Statement I and II both are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

187. **Statement-I :** Myometrium exhibits strong uterine contraction during delivery of the baby.
Statement-II : Myometrium is the muscular layer present in the uterus.
(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect
(3) Only statement-I is correct.
(4) Only statement-II is correct.

188. Foetal sex can be determined by examining cells from amniotic fluid by looking for :
(1) Barr bodies (2) Autosomes
(3) Chaismata (4) Kinetochore

189. **Statement-I :** Diaphragms, cervical caps and vaults are barriers made up of rubber and are non-reusable.
Statement-II : Hormonal IUCD make the uterus unsuitable for implantation and the cervix hostile to the sperm.
(1) Statement I and II both are correct
(2) Statement I and II both are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

190. Which is **not** a part of **ex-situ** conservation ?
(1) Gene bank (2) Botanical garden
(3) Tissue culture (4) Wild life sanctuary

191. In species - Area relationship the value of Z depends on which of the following factors ?
(a) Species richness
(b) Size of area
(c) Taxonomical category the species belongs
(1) (a) alone
(2) (b) alone
(3) (a) and (b)
(4) (a), (b) and (c)

192. Which of the following is not associated with increase in atmospheric concentration of CO_2 ?
(1) Increase in mean annual temperature
(2) Increase in sea level
(3) Expected increase in productivity of C_3 plants
(4) Increase in the rate of ozone depletion

193. Montreal protocol to protect stratospheric ozone was signed in Canada during which year :-
(1) 1731 (2) 1968 (3) 1987 (4) 1927

194. DDT is categorised under :-
(A) Non-biodegradable pollutants
(B) Primary pollutants
(C) Qualitative pollutants
(D) Prime contaminant for Eutrophication
(1) A only (2) B, C only
(3) A, B and C (4) A, B, C and D

195. Which of the following does **not** differ is *Nostoc* and *Chlorella* ?
(1) Cell membrane (2) Cell wall
(3) Ribsome (4) Nuclear organization

196. If a PMC has 20 pg DNA, then how much DNA will be present in pollen grains?
(1) 10 pg (2) 5 pg
(3) 20 pg (4) 15 pg

197. **Assertion :-** Careful analysis of records shows extinctions across taxa are not random.
Reason :- Amphibians are more vulnerable to extinction.
(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

198. If resources are unlimited then population growth curve follow which equation?
(1) $N_t = N_0 e^{rt}$
(2) $\frac{dN}{dt} = rN\left(1 - \frac{N}{k}\right)$
(3) $\frac{dN}{dt} = rN$
(4) Both (1) & (3)

199. Match column-I with column-II and choose the correct option.

	Column-I		Column-II
(A)	Nile perch in lake Victoria	(I)	Obvious reasons for biodiversity conservation
(B)	Narrowly utilitarian	(II)	Habitat destruction
(C)	Main cause for biodiversity loss	(III)	In-situ conservation
(D)	Hotspots	(IV)	Alien species

(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(2) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(3) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
(4) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

200. In the somatic cell cycle :-
(1) In G_1 phase, DNA content is double the amount of DNA present in the original cell.
(2) DNA replication takes place in S-phase.
(3) A short interphase is followed by a long mitotic phase.
(4) G_2 phase follows mitotic phase.

SPACE FOR ROUGH WORK

QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)

ENTHUSE ADV. COURSE : MEF

MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
06 13-08-2023	SYLLABUS COVERED FROM 17-07-2023 TO 12-08-2023	P	RAY OPTICS & OPTICAL INSTRUMENTS
		C	NOMENCLATURE, ISOMERISM
		B	LIFE CYCLE OF ANGIOSPERM & REPRODUCTION IN ORGANISM

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